

Member ID: _____

Time: _____

Rank: _____



NETWORK ADMINISTRATION USING CISCO (315) **NATIONAL 2023**

Total Points (50 @ 2 points each) _____ **(100 points)**

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. What is the purpose of LSA type 1 in OSPF?
 - A. It summarizes all networks within the OSPF area
 - B. It distributes the router ID and all connected networks of that router
 - C. It is used to redistribute routes from other routing protocols
 - D. It distributes the router ID
2. For OSPF, where are LSA type 1 packets distributed?
 - A. Within all OSPF areas
 - B. Only to directly connected OSPF routers
 - C. Only to the Area Border Router
 - D. Within the OSPF area where the LSA packet originates
3. What is a role of the Area Boarder Router in OSPF?
 - A. It is the default gateway for the hosts in the OSPF area
 - B. It is always the Designated Router of the OSPF area
 - C. It always runs other routing protocols to translate them into the OSPF area
 - D. It distributes summarized routes of the area to other OSPF areas
4. What interfaces would the OSPF command “network 192.168.12.0 0.0.0.255 area 0” effect?
 - A. All interfaces on the router that have an IP address in the range 192.168.12.0 to 192.168.12.255
 - B. All interfaces on the router that have an IP address outside the range 192.168.12.0 to 192.168.12.255
 - C. All interfaces on the router that have an IP address in the form of x.x.x.0 (X indicates wildcards)
 - D. All interfaces on the router that have an IP address in the form of x.x.x.0 to x.x.x.255 (X indicates wildcards)
5. Which of the following is required for an OSPF adjacency to form?
 - A. The interface must have the same MTU as the connected interface on the other router
 - B. The interface must be in the same IPv4 subnet as the connected interface on the other router
 - C. The interface must be in the same OSPF area as the interface of the other router
 - D. All of the other answers are correct.
6. What is a benefit of using VLANs
 - A. Connects segments of the network together
 - B. Enables the use of Access Control Lists
 - C. Separates traffic on the switch
 - D. Enabled the use of port security

7. Which command properly adds an access port to VLAN 20?
 - A. switch mode access
 - B. switchport mode access vlan 20
 - C. switchport access vlan 20
 - D. switchport vlan 20 access
8. With VLANs, what is the purpose of a trunk port?
 - A. Allows one port to send/receive traffic for one VLAN
 - B. Allows one port to send/receive tags for OSPF communication
 - C. Allows one port to send/receive traffic for multiple VLANs
 - D. Allows one port to send/receive traffic for multiple OSPF areas
9. What is the result of running the command “show vlan id 20” on a switch?
 - A. It will show the VLAN name as well as all ports that belong to the VLAN
 - B. It will show the VLAN name
 - C. It will show all the ports that belong to the VLAN
 - D. It will show all VLANs are able to communicate with VLAN 20
10. What commands set and enable an IP Address on a VLAN?
 - A. ip address 192.168.10.1 255.255.255.0, enable
 - B. address 192.168.10.1, no shutdown
 - C. address 192.168.10.1, enable
 - D. ip address 192.168.10.1 255.255.255.0, no shutdown
11. Which of the following addresses is in the same IPv4 subnet as 172.48.50.70/22
 - A. 172.48.51.30
 - B. 172.48.47.250
 - C. 172.48.53.25
 - D. 172.48.42.128
12. Which of the following addresses is in the same IPv4 subnet as 192.168.52.135/27
 - A. 192.168.52.126
 - B. 192.168.52.155
 - C. 192.168.52.10
 - D. 192.168.52.166
13. What is the broadcast for the IP address 192.168.72.113/26
 - A. 192.168.72.65
 - B. 192.168.72.127
 - C. 192.168.72.191
 - D. 192.168.72.255
14. What subnet mask could be applied to 172.16.0.0/16 to minimally create 170 networks?
 - A. 255.255.192.0
 - B. 255.255.240.0
 - C. 255.255.255.0
 - D. 255.255.254.0

15. Which of the following addresses can be assigned to an end device?
- A. 192.168.10.64/25
 - B. 192.168.10.192/27
 - C. 192.168.10.96/28
 - D. 192.168.10.127/29
16. What command enables IPv6 on Cisco devices?
- A. no ipv6 disable
 - B. ipv6 enable
 - C. no ipb6 unicast-routing disable
 - D. ipv6 unicast-routing
17. Which of the following is an IPv6 address?
- A. FD01::075:0004::0005
 - B. 32::0M
 - C. 3CD:88:909:F00D:33333::256
 - D. ::
18. What is the purpose of the Unique Local Address for IPv6?
- A. It allows internet communications
 - B. It allows intra-network communications
 - C. It replaces the link-local address for local communications
 - D. It is an IPv4 feature, not an IPv6 feature
19. Which of the following is an IPv6 Link-Local address?
- A. FE80:554::840:B:128
 - B. FE80:554:840:B::128
 - C. FE80::554:840:B:128
 - D. FE80:554:840:B:128
20. What IPv6 address would routers use to send routing information between one another?
- A. Unique Local Address
 - B. Global Unicast Address
 - C. Link-Local Address
 - D. Loopback Address
21. What is a reason to use Rapid Spanning-Tree over Spanning-Tree protocol?
- A. RSPT will reconverge faster in the case of a network change
 - B. Port states transition from one to another faster
 - C. Ports send information about their roles
 - D. All of the other answers are correct
22. What command enables Rapid Per-VLAN Spanning-Tree protocol on a Cisco switch?
- A. spanning-tree mode rapid-spanning-tree
 - B. spanning-tree mode rapid-pvst
 - C. spanning tree mode rapid-pvst
 - D. spanning tree mode rapid-spanning-tree

23. A host is not able to obtain an IP address when it is booted. It is determined that Spanning-tree is blocking the port before DHCP can give it an IP address. What is the best solution to eliminate the issue?
- A. Enable PortFast on the interface the host is connected to on the switch
 - B. Disable spanning-tree on the port
 - C. Disable spanning-tree
 - D. Set the port state transition time for spanning-tree to be lower
24. The command “show spanning-tree active” was ran on a switch and the following entry is found: “Gi0/1 Desg FWD 4 128.25 P2p”. What can be determined from this output?
- A. This switch connects to a downstream device
 - B. This switch port is being blocked on the downstream device
 - C. This switch is the root switch
 - D. This switch port is using fibre optic to connect to the other device
25. The command “show spanning-tree active” was ran on a switch and the following entry is found: “Gi0/2 Root FWD 4 128.26 P2p”. What can be determined from this output?
- A. This switch is the root switch
 - B. This switch port is used to connect downstream
 - C. This switch has rapid spanning-tree enabled
 - D. This switch port is used to connect upstream toward the root switch
26. A port on a switch is connected to a downstream switch. When a second hosts is connected to the downstream switch, after a moment, the port on the upstream switch becomes disabled. What is a cause of this?
- A. Port security is enabled on the port and only one MAC address is allowed
 - B. Spanning-Tree protocol is blocking the port
 - C. User authentication has not been enabled on that port
 - D. The switch does not have enough power to allow the port to work
27. When using port security, what is the purpose of the mode “restrict”?
- A. It blocks traffic from unknown sources while also notating the event by sending a message
 - B. It blocks traffic from unknown sources
 - C. It shuts down the port when a discrepancy is found
 - D. It allows traffic from unknow sources but notates the event by sending a message
28. What is the command to enable the port security mode “protect” on a port?
- A. switchport port-security mode protect
 - B. switchport port-security violation protect
 - C. switchport port-security protect
 - D. switchport protect port-security

29. After applying the port-security mode of shutdown on a port, it is not disabling itself when more than the allowed number of MAC addresses are found on it. Why is this the case?
 - A. The switch does not support the port-security feature
 - B. The administrator forgot to run the command “switchport port-security” on the port to activate port security
 - C. Port security applies to IP addresses, not MAC addressed
 - D. The administrator forgot to run the command “switchport port-security enable” on the port to activate port security

30. When running the command to see the status of port security on a port, it reports back that there have been 61 violations on it, but the port has not been disabled. What could be the cause of this?
 - A. Port security has not been enabled on the port
 - B. Port security only allows the display of violations. It does not affect the operations of the port
 - C. Port security has been enabled on the port and it is currently in restrict mode
 - D. Not enough violations have occurred to trigger the disabling of the port

31. Which entry below blocks source traffic specifically from hosts in the range 192.168.1.32 to 192.168.1.63?
 - A. access-list 101 deny ip 192.168.1.32 0.0.0.224 any
 - B. access-list 10 deny ip any 192.168.1.32 0.0.0.31
 - C. access-list 101 deny ip 192.168.1.32 0.0.0.31 any
 - D. access-list 10 deny ip 192.168.1.32 0.0.0.31 any

32. The following two commands were entered for an extended access control list: “access-list 101 permit ip any any”; “access-list 101 deny ip 192.168.20.64 0.0.0.31 any”. However, after applying this to a port in an appropriate direction, it is found that all traffic is being allowed. What is the cause?
 - A. The wildcard mask was set incorrectly
 - B. The access control list was created with the wrong identifying number
 - C. The commands use the keyword “any” rather than the keyword “all”
 - D. The general statement was made before the specific statement

33. Which entry below allows destination traffic specifically to hosts in the range 192.168.1.32 to 192.168.1.63?
 - A. access-list 101 permit ip any 192.168.1.32 0.0.0.224
 - B. access-list 101 permit ip any 192.168.1.32 0.0.0.31
 - C. access-list 10 permit ip 192.168.1.32 0.0.0.31 any
 - D. access-list 10 permit ip any 192.168.1.32 0.0.0.31

34. What entry below allows web traffic to the host 192.168.102.58?
 - A. access-list 101 permit tcp any 192.168.102.58 0.0.0.0 eq 80
 - B. access-list 101 permit tcp any 192.168.102.58 0.0.0.0 eq 85
 - C. access-list 101 permit tcp any 192.168.102.58 0.0.0.0 eq 90
 - D. access-list 101 permit tcp any 192.168.102.58 0.0.0.0 eq 95

35. What entry below allows SMTP traffic to the host 192.168.102.58?
- A. access-list 101 permit tcp any 192.168.102.58 eq 21
 - B. access-list 101 permit tcp any 192.168.102.58 eq 25
 - C. access-list 101 permit tcp any 192.168.102.58 eq 80
 - D. access-list 101 permit tcp any 192.168.102.58 eq 110
36. What TCP port is used for DHCP?
- A. Port 63
 - B. Port 53
 - C. DHCP does not use TCP
 - D. Port 67
37. What UDP port is used for DHCP?
- A. Port 67
 - B. Port 63
 - C. Port 53
 - D. DHCP does not use UDP
38. What commands below creates a DHCP Pool and gives it the network 192.168.20.0/26?
- A. ip dhcp new pool1; network 192.168.20.0 255.255.255.192
 - B. ip dhcp pool pool1; network 192.168.20.0 0.0.0.63
 - C. ip dhcp new pool1; network 192.168.20.0 0.0.0.192
 - D. ip dhcp pool pool1; network 192.168.20.0 255.255.255.192
39. For DHCP server to operate on a Cisco router, the router must have an interface with an IP address that exists in the DHCP pool.
- A. True
 - B. False
40. What commands are used to set a DHCP reservation range of 192.168.35.1 to 192.168.35.99?
- A. ip dhcp excluded-address 192.168.35.1 192.168.35.99
 - B. ip dhcp pool LAN; ip dhcp excluded-address 192.168.35.1 192.168.35.99
 - C. ip dhcp reservation 192.168.35.1 192.168.35.99
 - D. ip dhcp pool LAN; ip dhcp reservation 192.168.35.1 192.168.35.99
41. What port is commonly used by Domain Name Service?
- A. UDP 53
 - B. TCP 53
 - C. UDP 67
 - D. TCP 67

42. What best describes a Fully Qualified Domain Name?
- A. A common web address for large websites
 - B. An alpha numerical string with segments separated by periods that notate an object that belongs in a hierarchy
 - C. The hostname of a system
 - D. A string with segments that are separated by periods that notates an object and how it relates to other objects in its immediate network
43. For DNS, what is a reverse lookup?
- A. It is used to identify an IP Address from a FQDN
 - B. It is used to identify the server that holds the IP Address/DNS name mapping
 - C. It is used to identify other DNS servers that hold the IP Address/DNS name mapping
 - D. It is used when an DNS name is looked up by an IP Address
44. What is the purpose of NTP?
- A. It ensures proper packet size is used throughout the network
 - B. It allows DHCP to work in a timely manner
 - C. It decentralizes DNS so that DNS attacks are mitigated
 - D. It synchronizes clocks of networked devices
45. For NTP, what stratum is considered higher authority?
- A. Stratum 0
 - B. Stratum 1
 - C. Stratum 2
 - D. Stratum 3
46. What command below properly sets NTP to use the server 192.168.24.10 on a Cisco device?
- A. ntp server 192.168.24.10
 - B. ntp 192.168.24.10
 - C. set ntp 192.168.24.10
 - D. ntp 192.168.24.10 enable
47. When an Ethernet header is added to a data object, what is the object now referred to as?
- A. Packet
 - B. Frame
 - C. Segment
 - D. Bits
48. When the Ethernet layer is removed and addressing for an IPv4 environment is left, what is the data object referred to as?
- A. Frame
 - B. Segment
 - C. Packet
 - D. Bits

49. What ensures that both participants of a VPN connection are communicating with the correct encryption methods and keys?
- A. Authentication
 - B. Key Exchange
 - C. IOS Version
 - D. IPv6 Header
50. What is a common security appliance that can be installed into the network?
- A. Frame-Relay
 - B. IOS
 - C. Nexus
 - D. IPS